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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

GRADUATION THESIS

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HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

PDF-TO-DOCX CONVERSION AND PDF EDITING WEB-BASED APPLICATION

- MOTIVATION
- PROBLEM
- PROPOSED SOLUTION
- CONVERSION PIPELINE
- IMPLEMENTATION
- EXPERIMENTAL RESULTS
- CONTRIBUTIONS
- CONCLUSION

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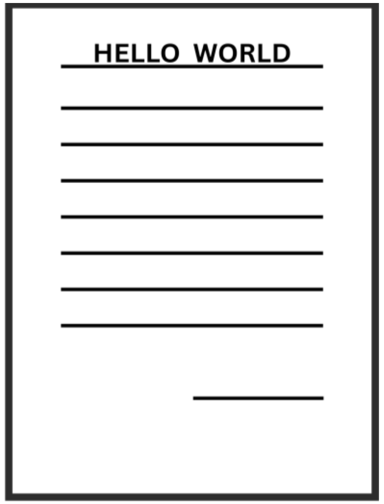
MOTIVATION

- PDF is widely used
- PDF is not editable
- Existing tools have limitations
- Accurate conversion remains challenging

Why PDF-to-DOCX is difficult?

- No paragraph information
- No reading order
- No semantic labels
- Absolute coordinates
- Complex structured elements

PDF Page



PDF Internal Structure

Raw text “HELLO WORLD”

Bbox = (121, 223, 497, 355)

Font = Times New Roman

Size = 12

Color = #000000

PROPOSED SOLUTION

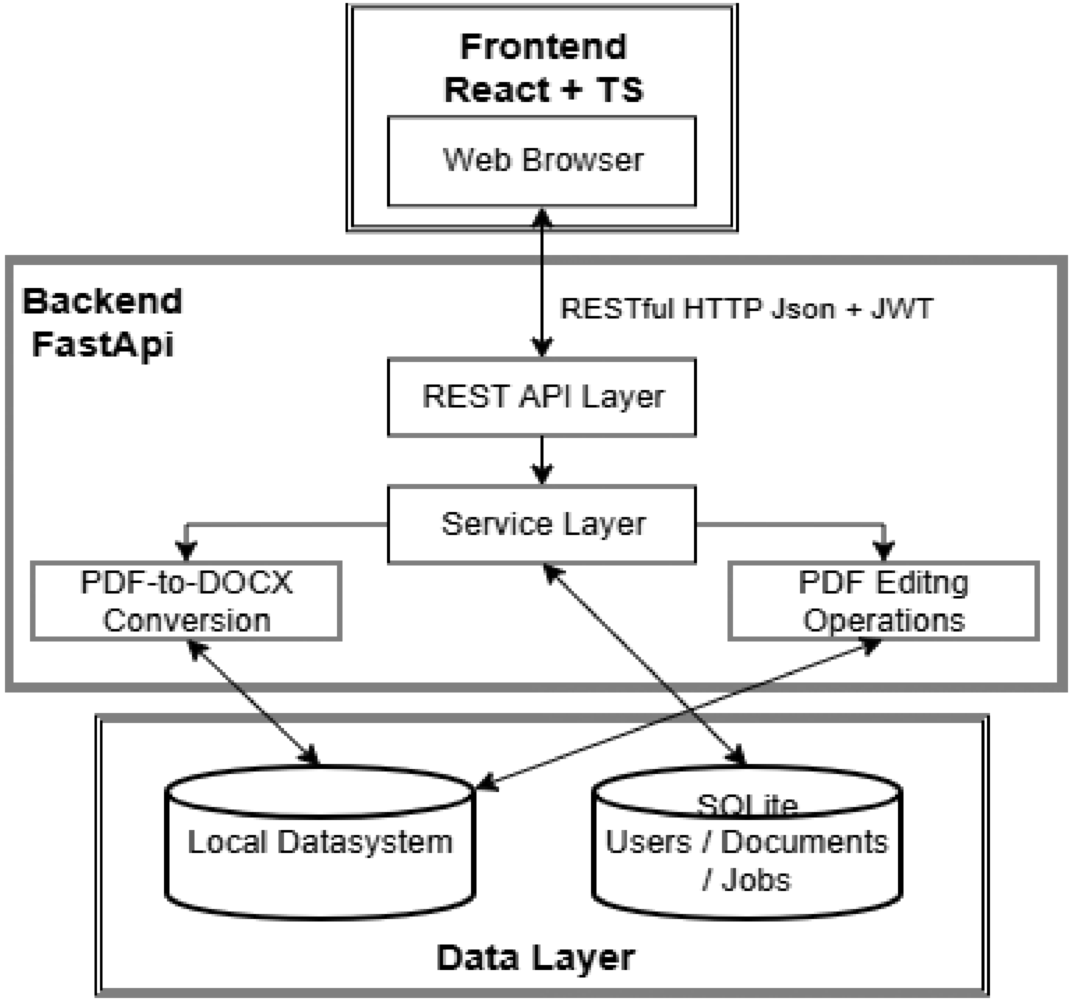
PDF → DOCX
Editable
Preserving layout

AutoDoc

PDF editing operations

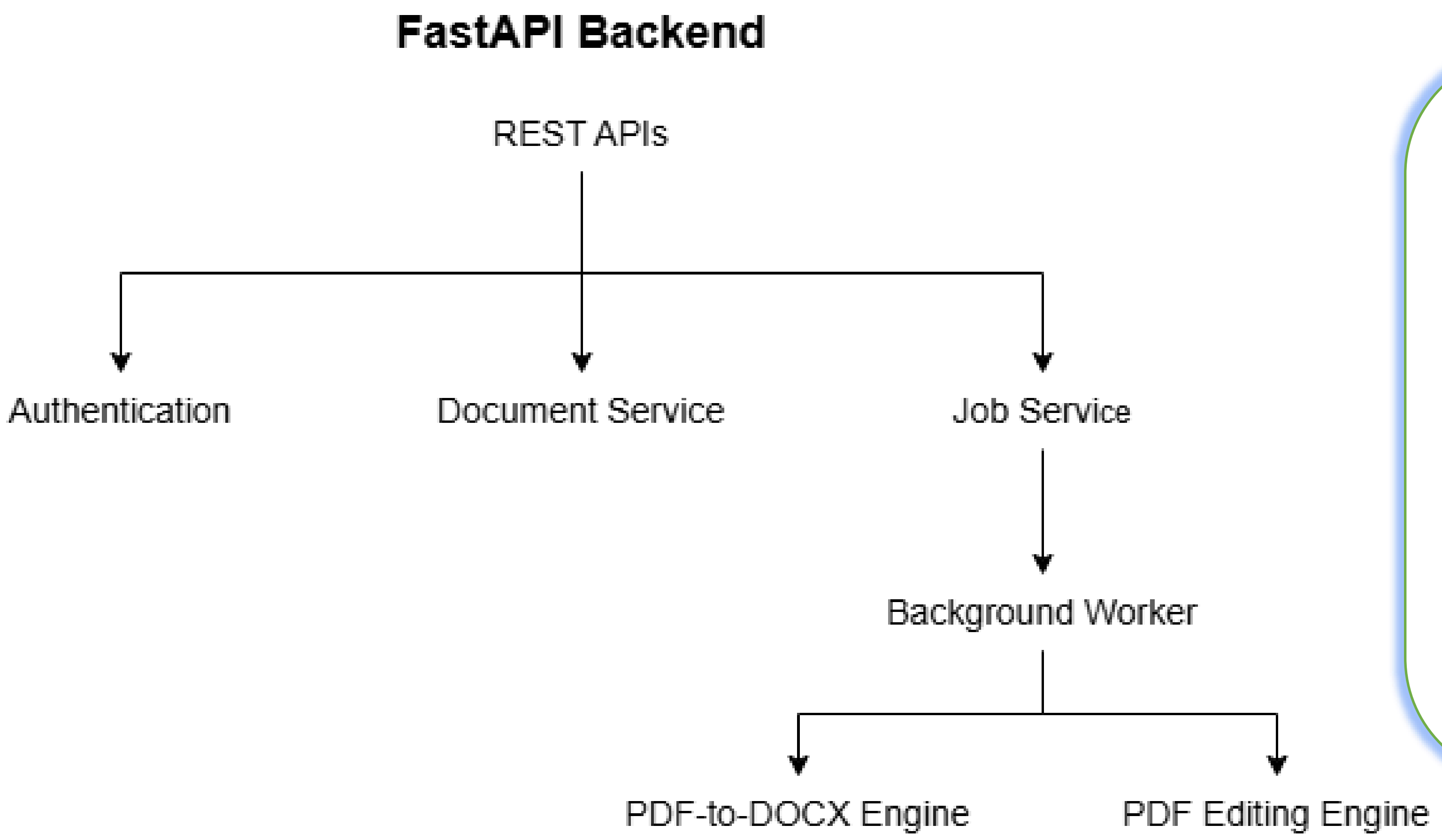
Core idea: Hybrid ML + Rule-based Heuristics

SYSTEM ARCHITECTURE



- Frontend, backend: Rest APIs
- Independent modules: PDF-DOCX Conversion, PDF Editing
- Conversion engine: ML + Rule-based heuristics
- Backend manages processing jobs
- Local storage

BACKEND DESIGN



Backend responsibilities:

- REST API requests
- Users, documents, processing jobs
- Execute document processing asynchronously
- Generated files for download

FRONTEND DESIGN

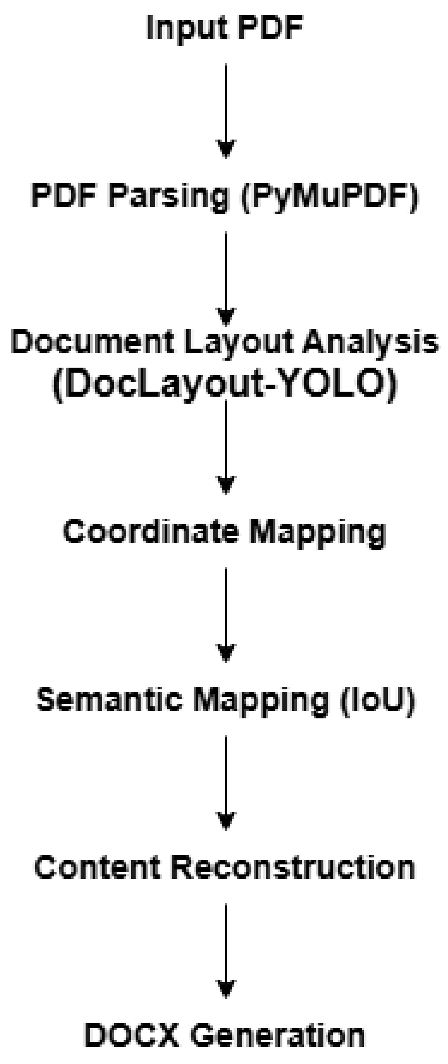
Main features:

- User authentication
- Document management
- PDF editing
- PDF-to-DOCX conversion
- Job monitoring
- Result preview & Download

Technologies:

- React
- TypeScript

PDF-TO-DOCX CONVERSION WORKFLOW



Main Processing Stages:

- PDF content extraction
- Layout analysis
- Coordinate mapping
- Semantic matching
- Content reconstruction
- DOCX generaion

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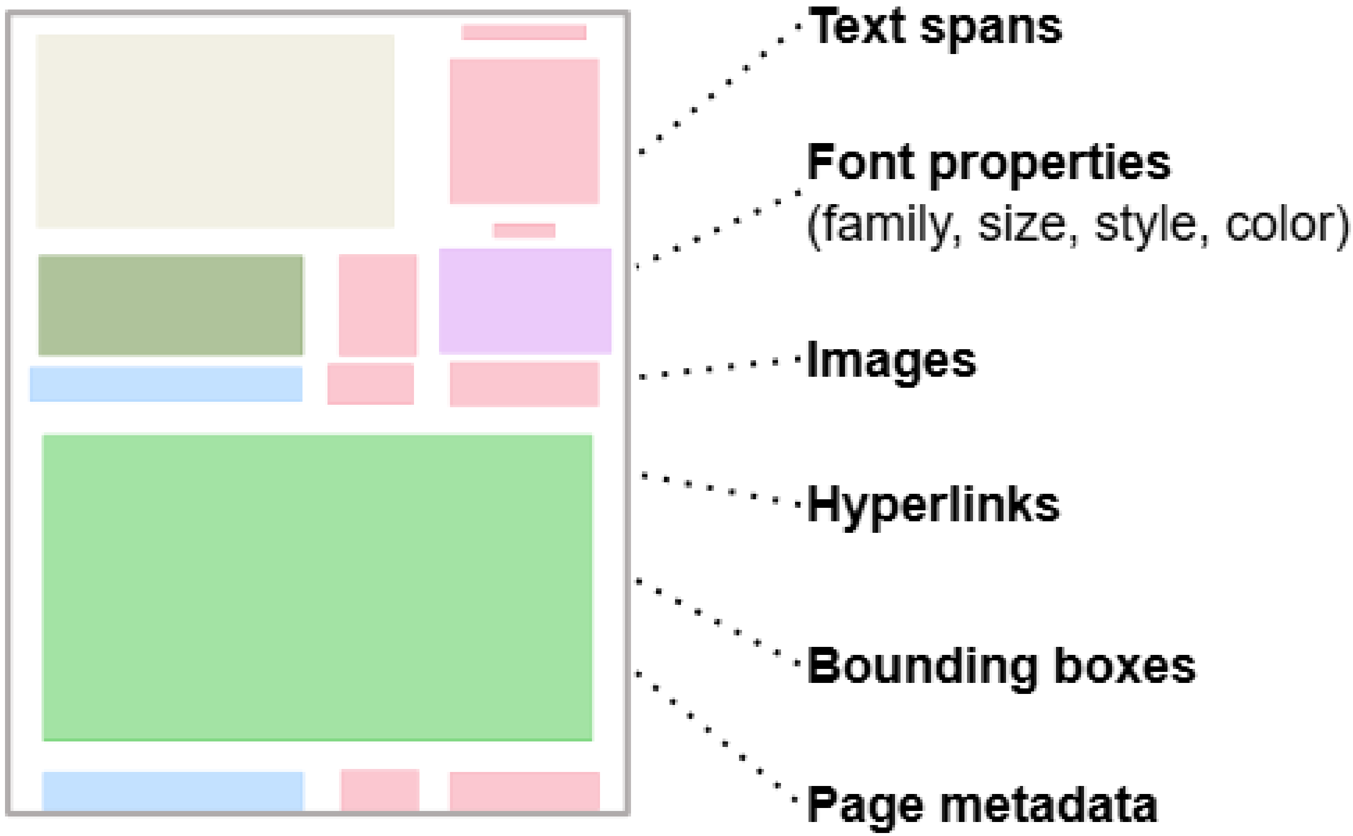
PDF CONTENT EXTRACTION



Why PyMuPDF?

- Direct extraction
- Preserve formatting
- Faster than OCR
- High accuracy for digital PDFs

Output: Structured PDF objects

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DOCUMENT LAYOUT

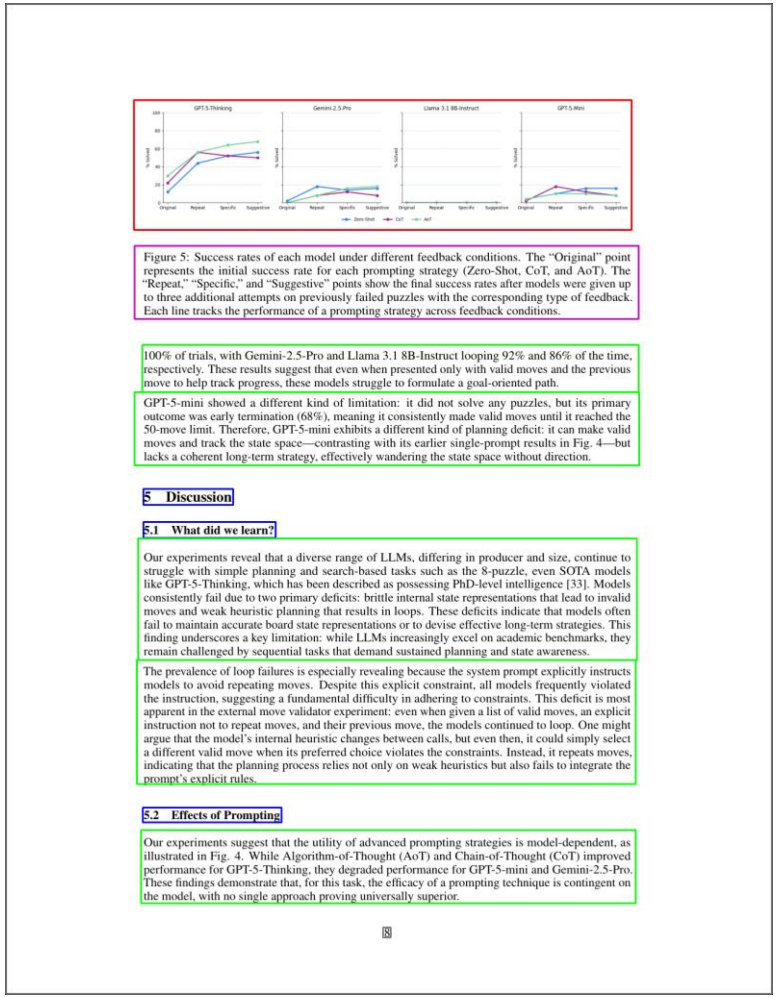

Figure 5: Success rates of each model under different feedback conditions. The "Original" point represents the initial success rate for each prompting strategy (Zero-Shot, CoT, and AoT). The "Repeat," "Specific," and "Suggestive" points show the final success rates after models were given up to three additional attempts on previously failed puzzles with the corresponding type of feedback. Each line tracks the performance of a prompting strategy across feedback conditions.

100% of trials, with Gemini-2.5-Pro and Llama 3.1 8B-Instruct looping 92% and 86% of the time, respectively. These results suggest that even when presented only with valid moves and the previous move to help track progress, these models struggle to formulate a goal-oriented path.

GPT-5-mini showed a different kind of limitation: it did not solve any puzzles, but its primary outcome was early termination (68%), meaning it consistently made valid moves until it reached the 50-move limit. Therefore, GPT-5-mini exhibits a different kind of planning deficit: it can make valid moves and track the state space—contrasting with its earlier single-prompt results in Fig. 4—but lacks a coherent long-term strategy, effectively wandering the state space without direction.

5 Discussion

5.1 What did we learn?

Our experiments reveal that a diverse range of LLMs, differing in producer and size, continue to struggle with simple planning and search-based tasks such as the 8-puzzle, even SOTA models like GPT-5-Thinking, which has been described as possessing PhD-level intelligence [33]. Models consistently fail due to two primary deficits: brittle internal state representations that lead to invalid moves and weak heuristic planning that results in loops. These deficits indicate that models often fail to maintain accurate board state representations or to devise effective long-term strategies. This finding underscores a key limitation: while LLMs increasingly excel on academic benchmarks, they remain challenged by sequential tasks that demand sustained planning and state awareness.

The prevalence of loop failures is especially revealing because the system prompt explicitly instructs models to avoid repeating moves. Despite this explicit constraint, all models frequently violated the instruction, suggesting a fundamental difficulty in adhering to constraints. This deficit is most apparent in the external move validator experiment: even when given a list of valid moves, an explicit instruction not to repeat moves, and their previous move, the models continued to loop. One might argue that the model's internal heuristic changes between calls, but even then, it could simply select a different valid move when its preferred choice violates the constraints. Instead, it repeats moves, indicating that the planning process relies not only on weak heuristics but also fails to integrate the prompt's explicit rules.

5.2 Effects of Prompting

Our experiments suggest that the utility of advanced prompting strategies is model-dependent, as illustrated in Fig. 4. While Algorithm-of-Thought (AoT) and Chain-of-Thought (CoT) improved performance for GPT-5-Thinking, they degraded performance for GPT-5-mini and Gemini-2.5-Pro. These findings demonstrate that, for this task, the efficacy of a prompting technique is contingent on the model, with no single approach proving universally superior.

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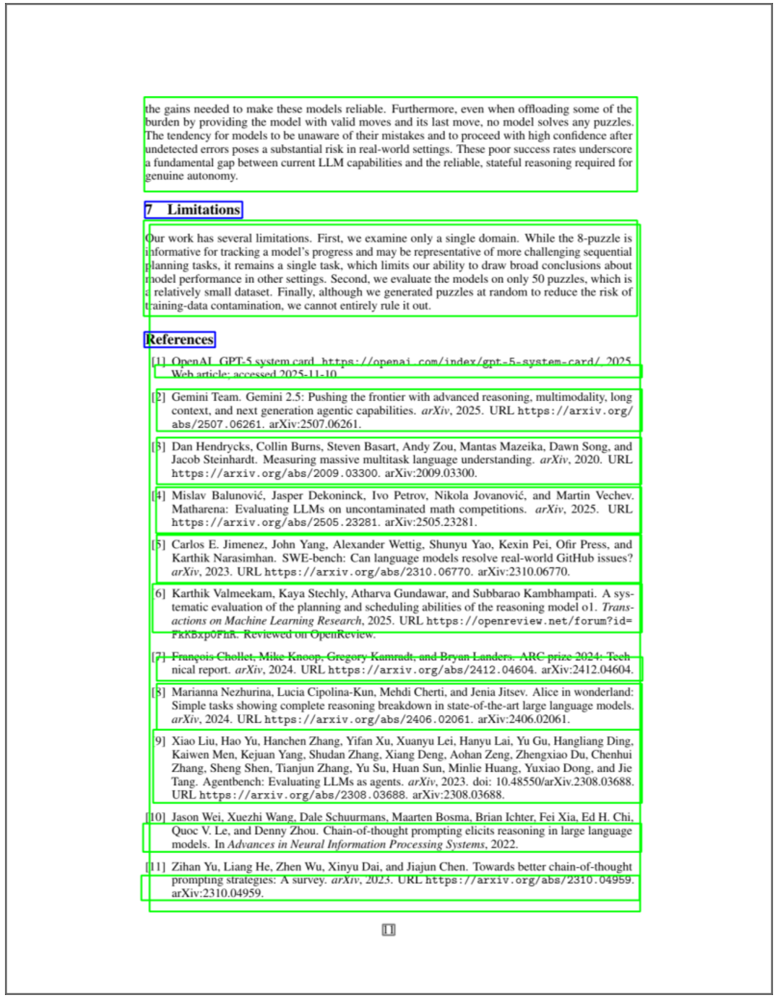

the gains needed to make these models reliable. Furthermore, even when offloading some of the burden by providing the model with valid moves and its last move, no model solves any puzzles. The tendency for models to be unaware of their mistakes and to proceed with high confidence after undetected errors poses a substantial risk in real-world settings. These poor success rates underscore a fundamental gap between current LLM capabilities and the reliable, stateful reasoning required for genuine autonomy.

7 Limitations

Our work has several limitations. First, we examine only a single domain. While the 8-puzzle is informative for tracking a model's progress and may be representative of more challenging sequential planning tasks, it remains a single task, which limits our ability to draw broad conclusions about model performance in other settings. Second, we evaluate the models on only 50 puzzles, which is a relatively small dataset. Finally, although we generated puzzles at random to reduce the risk of training-data contamination, we cannot entirely rule it out.

References

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11

DocLayout-YOLO

- Detect semantic regions
- Classify regions into categories
- Bouding boxes of detected regions

11 classes:

- Title
- Text
- Table
- Figure caption
- Figure
- Table caption
- Others

COORDINATE MAPPING

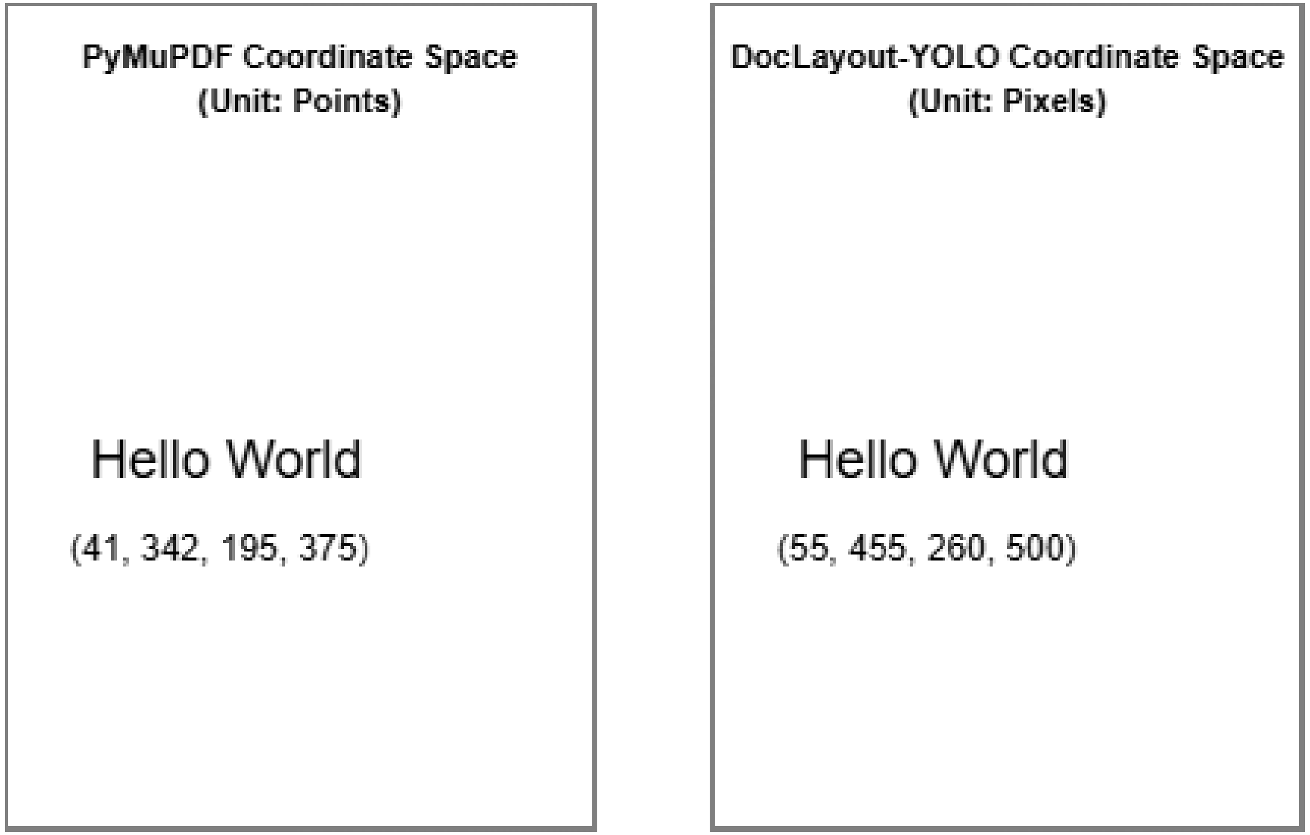
Problem:

Different coordinate systems:

- PyMuPDF: points
- DocLayout-YOLO: pixels

Solution:

- Convert PDF bounding boxes into image coordinates
- All object in a unified coordinate space



1 pt = 1.33px

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SEMANTIC MATCHING

PyMuPDF

DocLayout-YOLO

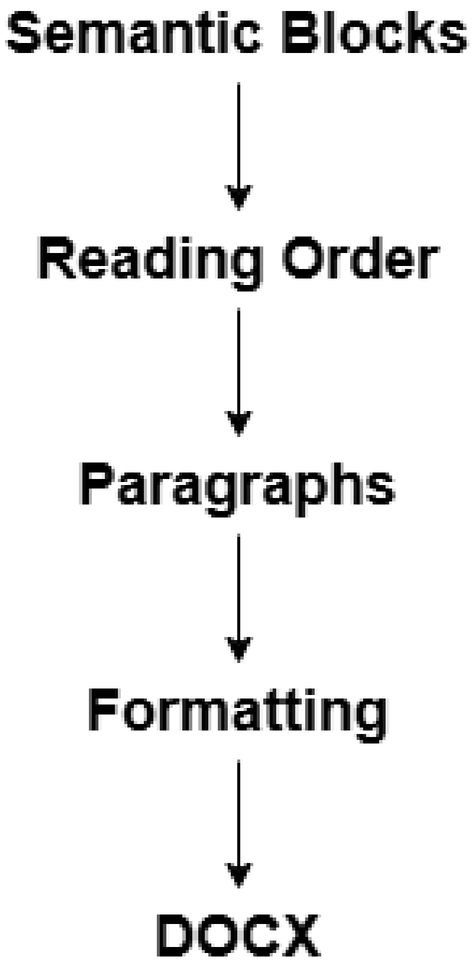
Title

Matching strategy:

- Compare mapped elements with detected layout regions
- Measure overlap = IoU
- Assign semantic label to best matching

$$IoU = \frac{Intersection\ Area}{Union\ Area}$$

DOCUMENT RECONSTRUCTION



Reconstruction Process:

- Recover logical reading order
- Merge related text spans into paragraphs
- Preserve formatting
- Convert document elements into DOCX objects

RECONSTRUCTION

Structural Refinement:

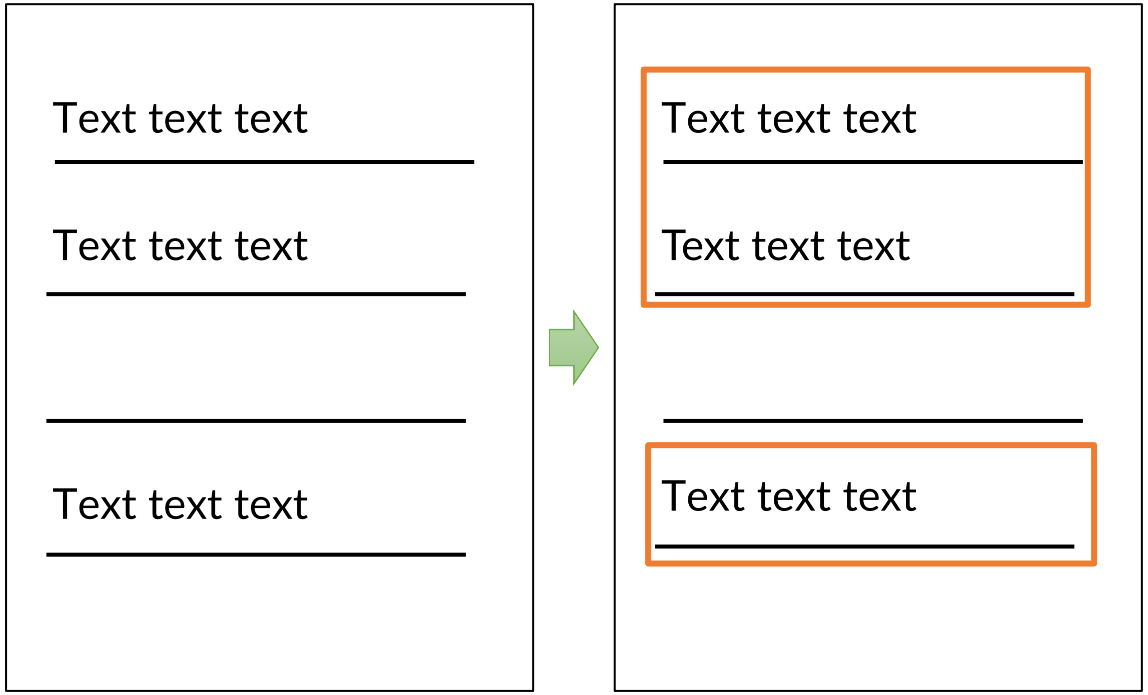
- Tight bounding box refinement
- Paragraph grouping
- Cross-page section continuity
- IoU & Containment ratio elements tracking

Content Refinement:

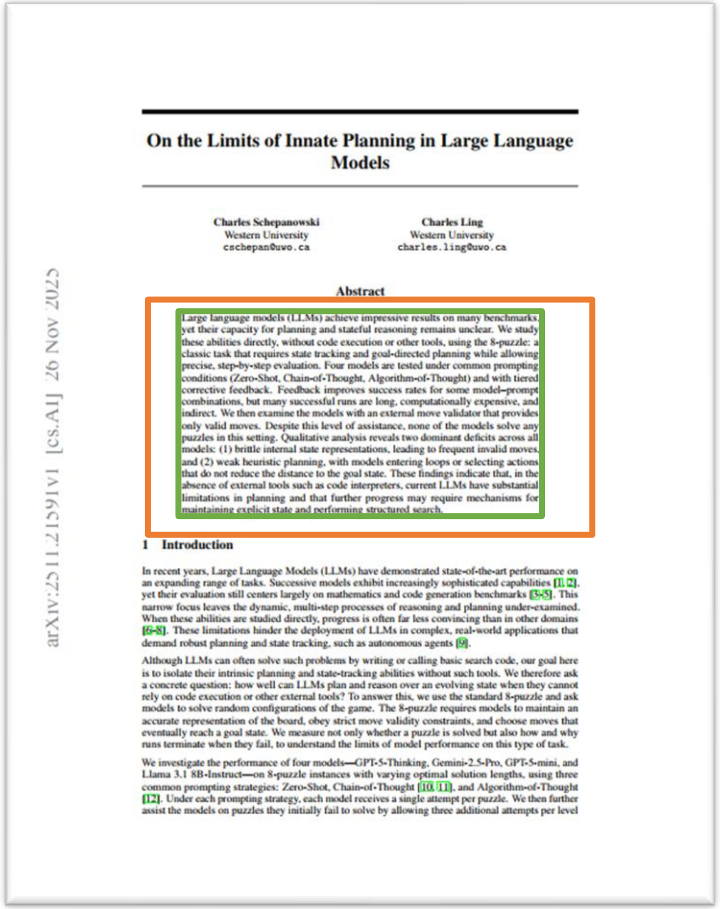
- External hyperlinks preservation
- Text properties preservation

RECONSTRUCTION

Paragraph grouping



Tight bounding box



arXiv:2511.21591v1 [cs.AI] 26 Nov 2025

On the Limits of Innate Planning in Large Language Models

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Abstract

Large language models (LLMs) achieve impressive results on many benchmarks, yet their capacity for planning and stateful reasoning remains unclear. We study these abilities directly, without code execution or other tools, using the 8-puzzle: a classic task that requires state tracking and goal-directed planning while allowing precise, step-by-step evaluation. Four models are tested under common prompting conditions (Zero-Shot, Chain-of-Thought, Algorithm-of-Thought) and with tiered corrective feedback. Feedback improves success rates for some model–prompt combinations, but many successful runs are long, computationally expensive, and indirect. We then examine the models with an external move validator that provides only valid moves. Despite this level of assistance, none of the models solve any puzzles in this setting. Qualitative analysis reveals two dominant deficits across all models: (1) brittle internal state representations, leading to frequent invalid moves, and (2) weak heuristic planning, with models entering loops or selecting actions that do not reduce the distance to the goal state. These findings indicate that, in the absence of external tools such as code interpreters, current LLMs have substantial limitations in planning and that further progress may require mechanisms for maintaining explicit state and performing structured search.

1 Introduction

In recent years, Large Language Models (LLMs) have demonstrated state-of-the-art performance on an expanding range of tasks. Successive models exhibit increasingly sophisticated capabilities [1, 2], yet their evaluation still centers largely on mathematics and code generation benchmarks [3–5]. This narrow focus leaves the dynamic, multi-step processes of reasoning and planning under-examined. When these abilities are studied directly, progress is often far less convincing than in other domains [6–8]. These limitations hinder the deployment of LLMs in complex, real-world applications that demand robust planning and state tracking, such as autonomous agents [9].

Although LLMs can often solve such problems by writing or calling basic search code, our goal here is to isolate their intrinsic planning and state-tracking abilities without such tools. We therefore ask a concrete question: how well can LLMs plan and reason over an evolving state when they cannot rely on code execution or other external tools? To answer this, we use the standard 8-puzzle and ask models to solve random configurations of the game. The 8-puzzle requires models to maintain an accurate representation of the board, obey strict move validity constraints, and choose moves that eventually reach a goal state. We measure not only whether a puzzle is solved but also how and why runs terminate when they fail, to understand the limits of model performance on this type of task.

We investigate the performance of four models—GPT-5-Thinking, Gemini-2.5-Pro, GPT-5-mini, and Llama 3.1 8B-Instruct—on 8-puzzle instances with varying optimal solution lengths, using three common prompting strategies: Zero-Shot, Chain-of-Thought [10, 11], and Algorithm-of-Thought [12]. Under each prompting strategy, each model receives a single attempt per puzzle. We then further assist the models on puzzles they initially fail to solve by allowing three additional attempts per level

RECONSTRUCTION

Cross-page section continuity

Adobe Acrobat

6 Conclusion

In this work, we examined a range of LLMs using the 8-puzzle to probe their planning and stateful reasoning capabilities. Our findings show that both SOTA and small models still struggle significantly with this seemingly simple task, with failures indicating critical deficits in state tracking and heuristic planning. While feedback and prompt engineering offer some help, they are costly and do not provide

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AutoDoc

the gains needed to make these models reliable. Furthermore, even when offloading some of the burden by providing the model with valid moves and its last move, no model solves any puzzles. The tendency for models to be unaware of their mistakes and to proceed with high confidence after undetected errors poses a substantial risk in real-world settings. These poor success rates underscore a fundamental gap between current LLM capabilities and the reliable, stateful reasoning required for genuine autonomy.

7 Limitations

Our work has several limitations. First, we examine only a single domain. While the 8-puzzle is informative for tracking a model's progress and may be representative of more challenging sequential planning tasks, it remains a single task, which limits our ability to draw broad conclusions about

IoU & Containment ratio tracking

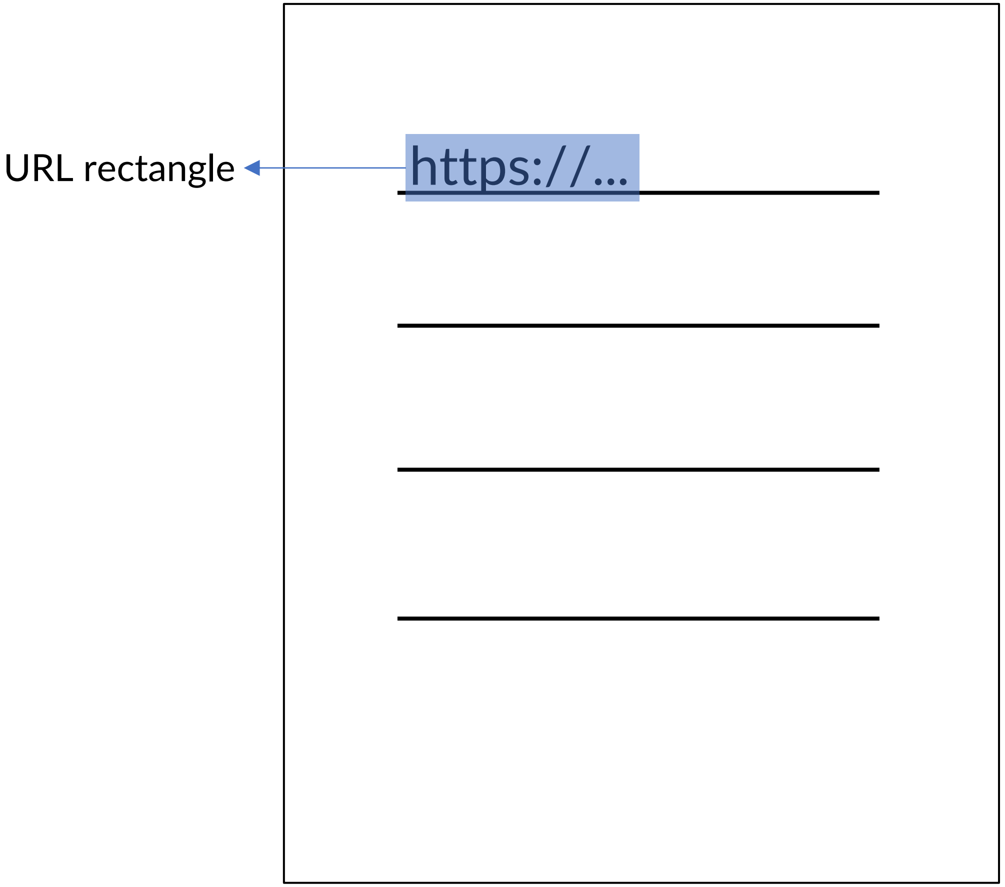
Model	Approach	Feedback	Solved N	Time (min)	Tokens	Attempts	Moves	Optimal
GPT-5-Thinking	AoT	Suggestive	34	23.92	75284	1.97	48.91	21.03
Gemini-2.5-Pro	Base	Repeat	9	6.74	58651	2.67	30.56	19.67
Llama 3.1 8B-Instruct	Base	Repeat	0	0	0	0	0	0
GPT-5-mini	CoT	Repeat	9	9.91	44272	2.67	28.44	19.11

Model	Approach	Feedback	Solved N	Time (min)	Tokens	Attempts	Moves	Optimal
GPT-5-Thinking	AoT	Suggestive	34	23.92	75284	1.97	48.91	21.03
Gemini-2.5-Pro	Base	Repeat	9	6.74	58651	2.67	30.56	19.67
Llama 3.1 8B-Instruct	Base	Repeat	0	0	0	0	0	0
GPT-5-mini	CoT	Repeat	9	9.91	44272	2.67	28.44	19.11

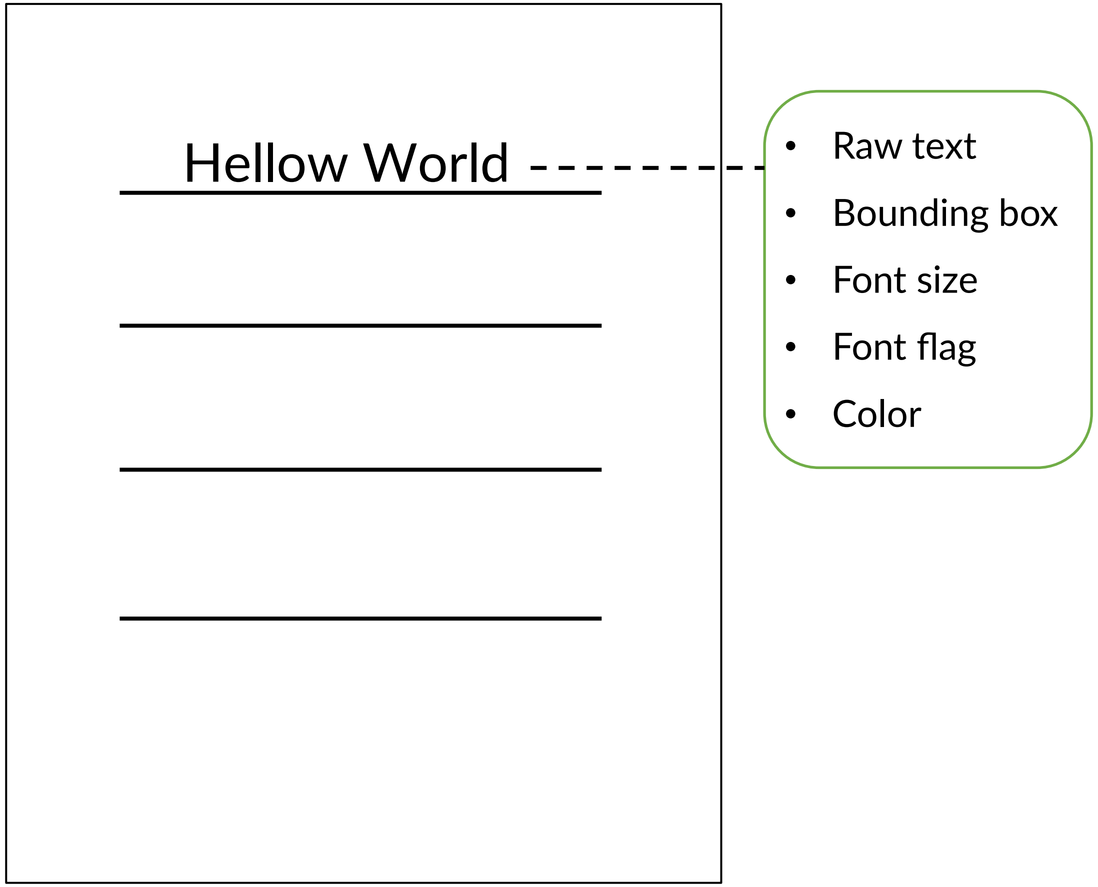
Table 1: Solved count and average time (minutes), tokens used, attempts, moves, and optimal moves for each model under its best-performing condition. We broke ties for GPT-5-mini and Gemini-2.5-Pro by choosing the approach that required the fewest tokens.

RECONSTRUCTION

External hyperlinks preservation



Text properties preservation



DOCX GENRATION

Structured Elements

↓

Paragraph

↓

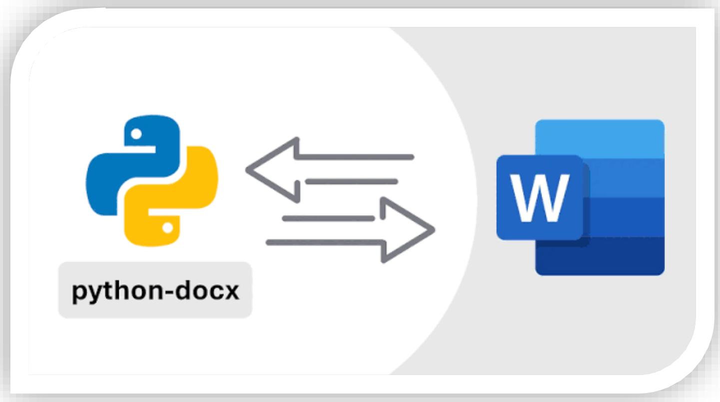
Run

↓

Image

↓

DOCX File

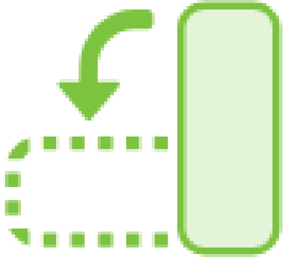


Editable Microsoft Word Document

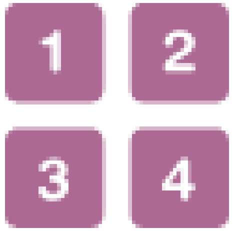
PDF EDITING FEATURES

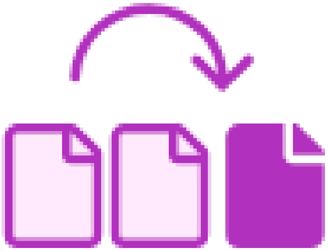
Merge PDF


Split PDF


Organize PDF


Insert pages


Number pages


Reorder pages


Crop pages


DEMO

CONVERSION RESULT COMPARISON

Feature	AutoDoc	Adobe Acrobat	iLovePDF
Layout	✓	✓	✓
Font properties	✓	✓	✓
Images	✓	✓	✓
Editable tables	✗	✓	✓
Formulas	✗	✗	✗
Section continuity	✓	✗	✗
Internal hyperlinks	✗	✓	✓
External hyperlinks	✓	✓	✓
Multi-column	✓	✓	✓
Speed	✗	✓	✓

CONTRIBUTIONS

Hybrid PDF-to-DOCX Pipeline

- DocLyout-YOLO + Rule-based reconstruction

Semantic Document Reconstruction

- Logical document structure
- Section continuity across page boundaries

Reconstruction Refinement

- Improve layout quality
- Preserve hyperlinks, text format

Practical Conversion Quality

- Editable DOCX with high layout fidelity
- Comparable results with commercial tools

LIMITATIONS & FUTURE WORK

⚠️ Limitations

- "Text swallowing" due to AI imprecision
- Pagination discrepancy
- Unprocessed internal hyperlinks
- Section header without numeric prefixes
- Non-editable complex elements

🚀 Future work

- Support internal hyperlinks
- Optimize processing performance
- Reconstruct editable tables, formulas

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CONCLUSION

Project summary

- AutoDoc, a web-based PDF processing system
- Hybrid PDF-to-DOCX conversion pipeline, ML + rule-based heuristics
- PDF editing and document conversion
- Preserve document layout, formatting, hyperllinks
- Comaprable conversion quality with commercial tools

AutoDoc

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Thank you for your attention!